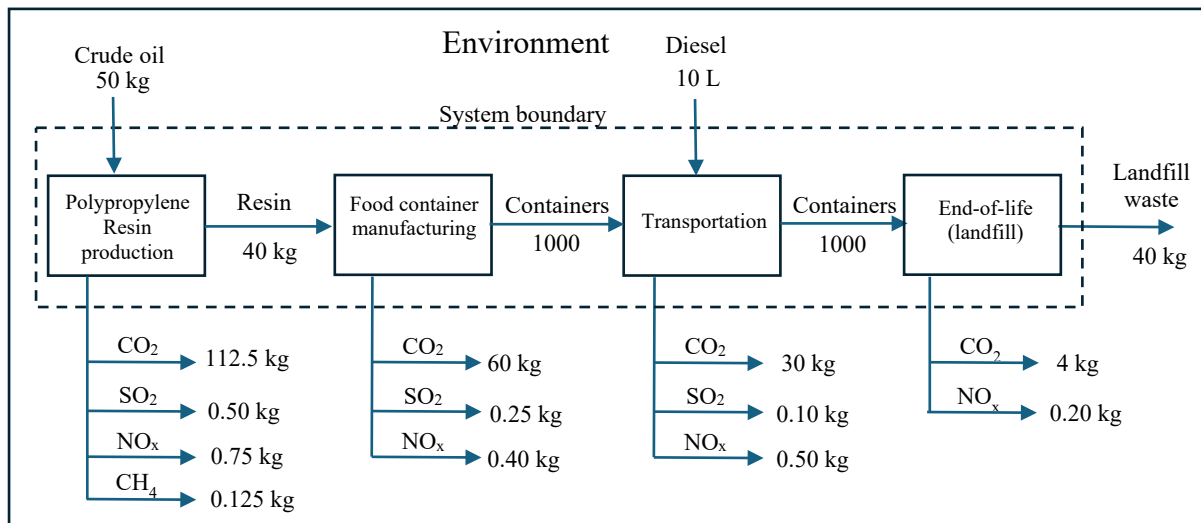


DSS5202 Sustainable Systems Analysis

Solutions to Assignment 1

1. Product System Diagram with reference flow quantities



2. Scaling factor computations

The scaling factors required to produce 1000 1L food containers can be determined by working backward from the functional unit reference flow:

Food Container Manufacturing

- Number of containers required = 1000
- Scaling factor = $1000/200 = 5$

	Input		Quantity	Ref flow
1	Polypropylene resin	kg	8	40
	Output			
1	1L Food containers	unit	200	1000
2	CO ₂	kg	12	60
3	SO ₂	kg	0.05	0.25
4	NO _x	kg	0.08	0.40
5	CH ₄ (Fossil methane)	kg	0	0

Polypropylene Resin Production

- Polypropylene resin = 40 kg.
- Scaling factor = $40 \text{ kg}/16 \text{ kg} = 2.50$

	Input		Quantity	Ref flow
1	Crude oil	kg	20	50
	Output			
1	Polypropylene resin	kg	16	40
2	CO ₂	kg	45	112.500
3	SO ₂	kg	0.20	0.500
4	NO _x	kg	0.30	0.750
5	CH ₄ (Fossil methane)	kg	0.05	0.125

Transportation

- Unit weight = 0.200 kg/container
- Total weight transported = $0.200 \times 1000 \times 10^{-3} = 0.200$ tons
- Distance transported = 500 km
- Scaling factor = $0.2 \times 500 / 10 = 10$

	Input		Quantity	Ref flow
1	Transport service	t.km	10	100
2	Diesel	L	1	10
	Output			
1	CO2	kg	3	30
2	SO2	kg	0.01	0.10
3	NOx	kg	0.05	0.50
4	CH4 (Fossil methane)	kg	0	0

End-of-Life (Landfill)

- Scaling factor = $1000/50 = 20$

	Input		Quantity	Ref flow
1	Used food containers	unit	50	1000
	Output			
1	Landfill waste	kg	2	40
2	CO2	kg	0.2	4.00
3	SO2	kg	0	0.00
4	NOx	kg	0.01	0.20
5	CH4 (Fossil methane)	kg	0	0

The unit process scaling factors are:

1	Polypropylene Resin Production	2.5
2	Food Container Manufacturing	5
3	Transportation	10
4	End-of-Life (Landfill)	20

3. Life Cycle Inventory Analysis

Product Flows

1	Polypropylene resin	40 kg
2	Food containers	1000

Elementary Flows

	Elementary flow		Polypropylene Resin Production	Food Container Manufacturing	Transportation	End-of-Life (Landfill)	Total LC Inventory
1	Crude oil extracted		50	0	0	0	50
2	Landfill waste		0	0	0	40	40
1	CO ₂	kg	112.5	60.0	30.0	4.000	206.5
2	SO ₂	kg	0.500	0.250	0.100	0	0.85
3	NO _x	kg	0.750	0.400	0.500	0.200	1.85
4	CH ₄ (Fossil methane)	kg	0.125	0	0	0	0.125

4. Life Cycle Midpoints Impact Analysis

Let the midpoints impact categories characterization factors matrix be denoted by

$$CF_{\text{midpoint}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 36 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0.36 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The scores for each midpoint impact category for each production process may be computed by matrix multiplication:

$$\begin{bmatrix} GWP100 \\ HOF \\ AP \\ EOF \\ FFP \end{bmatrix} = CF_{\text{midpoint}} \begin{bmatrix} 50 & 0 & 0 & 0 \\ 0 & 0 & 0 & 40 \\ 112.5 & 60 & 30 & 4 \\ 0.50 & 0.25 & 0.1 & 0 \\ 0.75 & 0.40 & 0.5 & 0.2 \\ 0.125 & 0 & 0 & 0 \end{bmatrix}$$

The midpoint category impact scores are given in the table below:

	Midpoint category		Polypropylene Resin Production	Food Container Manufacturing	Transportation	End-of-Life (Landfill)	Total impact score
1	GWP100	kg CO ₂ -eq	117.00	60.00	30.00	4.00	211.00
2	HOFP	kg NO _x -eq	0.750	0.400	0.500	0.200	1.850
3	AP	kg SO ₂ -eq	0.770	0.394	0.280	0.072	1.516
4	EOFP	kg NO _x -eq	0.750	0.400	0.500	0.200	1.850
5	FFP	kg oil-eq	50.000	0.000	0.000	0.000	50.000

5. Life Cycle Endpoints Impact Analysis

Let the midpoints-to-endpoint areas of protection characterization factors matrix be

$$CF_{end} = \begin{bmatrix} 9.28E-07 & 9.10E-07 & 0 & 0 & 0 \\ 2.80E-09 & 0 & 2.12E-07 & 1.29E-07 & 0 \\ 7.65E-14 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4.566E-01 \end{bmatrix}$$

The scores for the end impact category for each production process may be computed by matrix multiplication:

$$\begin{bmatrix} \text{Human Health} \\ \text{Terrestrial ecosystems} \\ \text{Freshwater ecosystems} \\ \text{Resource} \end{bmatrix} = CF_{end} \begin{bmatrix} 117 & 60 & 30 & 4 \\ 0.75 & 0.400 & 0.500 & 0.200 \\ 0.77 & 0.394 & 0.280 & 0.072 \\ 0.75 & 0.400 & 0.500 & 0.200 \\ 50 & 0 & 0 & 0 \end{bmatrix}$$

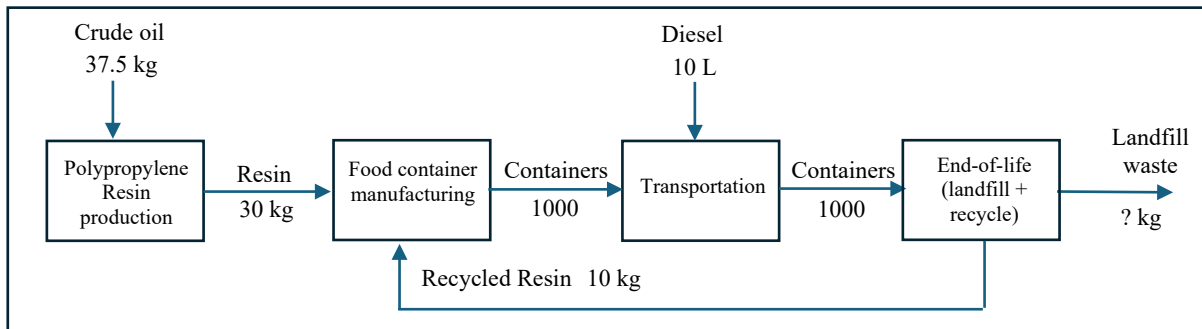
The endpoint impact scores are given in the table below:

	Area of protection	Unit	Polypropylene Resin Production	Food Container Manufacturing	Transportation	End-of-Life (Landfill)	Total impact score
1	Human health	DALY	1.09E-04	5.60E-05	2.83E-05	3.89E-06	1.9749E-04
2	Terrestrial ecosystems	Species.year	5.88E-07	3.03E-07	2.08E-07	5.23E-08	1.1508E-06
3	Freshwater ecosystems	Species.year	8.9505E-12	4.59E-12	2.295E-12	3.06E-13	1.6142E-11
4	Resource	USD2013	2.28E+01	0	0	0.00E+00	2.2830E+01

6. Product system with simple backflow

Given that 0.5 kg of resin is recycled back to container manufacturing for every 50 used containers, we recompute the new product reference flows via backward tracing.

New reference flows (emissions to air omitted)



The new scaling factor for each unit process can be computed as follows:

End-of-life management (Landfill + Recycle):

- Number of used containers = 1000.
- Amount of resin recycled = $1000/50 \times 0.5 = 10$ kg
- The scaling factor remains the same = $1000/50 = 20$.

Food container manufacturing:

- Total amount of resin needed to produce 1000 containers = 40
- Amount of virgin resin needed = $40 - 10 = 30$ kg
- The scaling factor remains the same as $1000/200 = 5$

Polypropylene Resin Production:

- Reference flow for virgin resin output = 30 kg (reduced from previous 40 kg)
- New scaling factor = $30/16 = 1.875$

Transportation

- The scaling factor remains the same as $100/10 = 10$.

New scaling factors

1	Polypropylene Resin Production	1.875
2	Food Container Manufacturing	5
3	Transportation	10
4	End-of-Life (Landfill)	20

Intuitively, less virgin resin needs to be produced now to satisfy the same functional unit. The other processes are unaffected.